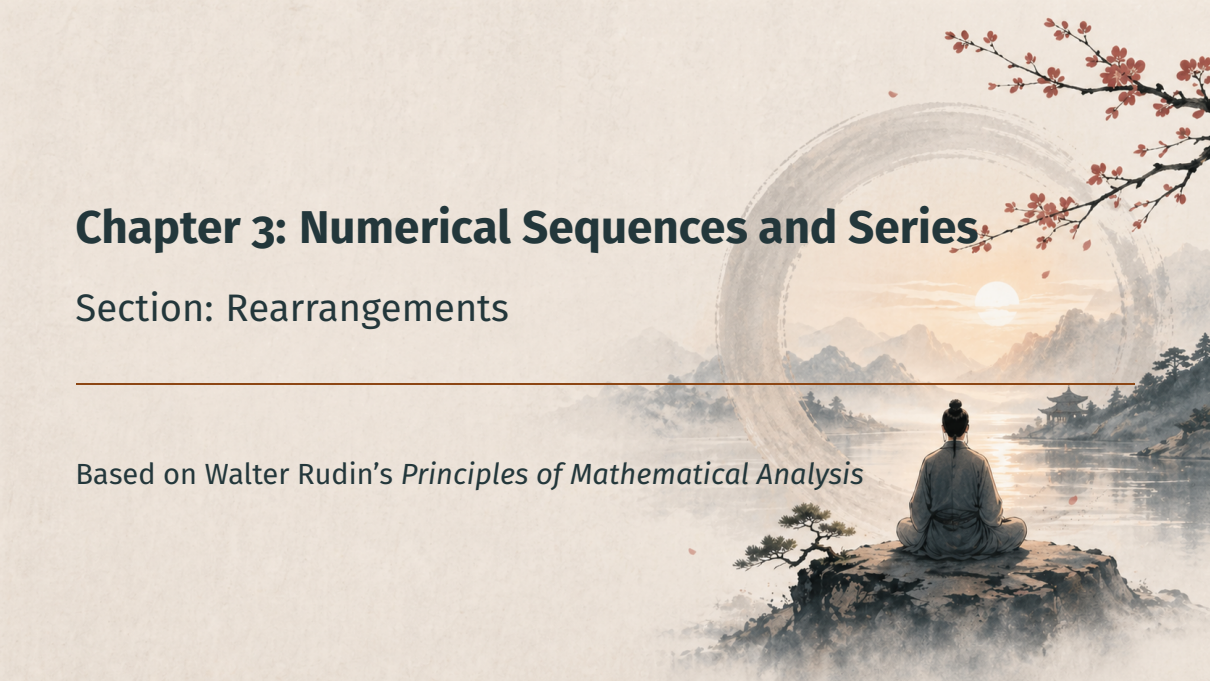


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person in a grey robe is seen from behind, sitting in a meditative pose on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains are visible. A large, faint circular frame, resembling a ring or a portal, is centered in the background. A small pavilion with a traditional roof sits on the right bank of the lake. A branch with red blossoms, likely cherry blossoms, extends from the top right corner into the frame. The overall atmosphere is serene and contemplative.

Section: Rearrangements

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

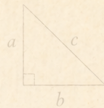
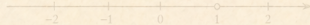
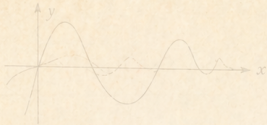
What Is a Rearrangement?

A Cautionary Example

Riemann's Rearrangement Theorem

Absolute Convergence and Rearrangements

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Does the Order of Summation Matter?

- For **finite** sums, order never matters: $a + b + c = c + a + b$
(commutativity + associativity)



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Does the Order of Summation Matter?

- For **finite** sums, order never matters: $a + b + c = c + a + b$ (commutativity + associativity)
- An infinite series is *not* just a set of numbers:

$$\sum a_n = \lim_{N \rightarrow \infty} s_N, \quad s_N = a_1 + a_2 + \cdots + a_N$$

The **order of the terms** determines the partial sums!

Does the Order of Summation Matter?

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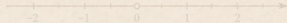
The **order of the terms** determines the partial sums!

- Do reordered series still **converge**? If so, to the **same sum**?

$$f(x) = \sin x$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

What Is a Rearrangement?



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Definition 3.52: Rearrangement

Definition 3.52

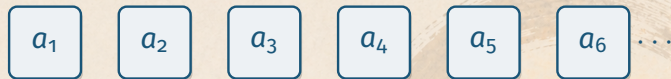
Let $\{k_n\}$, $n = 1, 2, 3, \dots$, be a sequence in which every positive integer appears **once and only once** ($\{k_n\}$ is a 1-1 function from J onto J ; see Definition 2.2). Putting

$$a'_n = a_{k_n} \quad (n = 1, 2, 3, \dots),$$

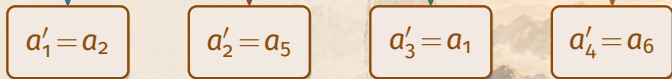
we say that $\Sigma a'_n$ is a **rearrangement** of Σa_n .

Definition 3.52: Rearrangement

original Σa_n



rearranged $\Sigma a'_n$



$\{k_n\} = (2, 5, 1, 6, \dots)$ is a bijection of J onto itself — a **permutation** of the same terms.

Same terms, new order $\Rightarrow$ partial sums $\{s_n\}$ and $\{s'_n\}$ consist of **entirely different numbers** in general.

A Cautionary Example



Example 3.53: Reordering the Alternating Harmonic Series

Example 3.53

The alternating harmonic series converges (Theorem 3.43):

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots \quad (22)$$

Consider the rearrangement (*two positive terms, then one negative*):

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots \quad (23)$$

Example 3.53: Reordering the Alternating Harmonic Series

Step 1: The original sum s satisfies $s < \frac{5}{6}$.

$$s = \underbrace{1 - \frac{1}{2} + \frac{1}{3}}_{=5/6} - \underbrace{\frac{1}{4} + \frac{1}{5}}_{<0} - \underbrace{\frac{1}{6} + \frac{1}{7}}_{<0} - \dots \implies \boxed{s < 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}}$$

Example 3.53: Reordering the Alternating Harmonic Series

Step 2: The rearranged partial sums s'_{3k} increase.

Group (23) into triples; the k th triple is

$$\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} > 0 \quad (k \geq 1),$$

so

$$s'_3 < s'_6 < s'_9 < \cdots, \quad s'_3 = 1 + \frac{1}{3} - \frac{1}{2} = \frac{5}{6}.$$

Example 3.53: Reordering the Alternating Harmonic Series

Step 3: Conclusion.

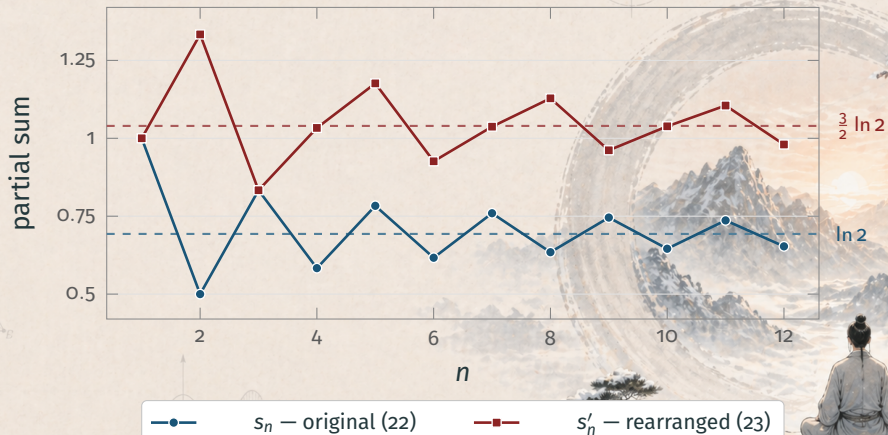
$$\limsup_{n \rightarrow \infty} s'_n > s'_3 = \frac{5}{6} > s$$

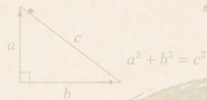
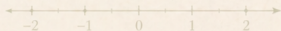
The rearrangement changes the sum

Series (23) does **not** converge to s .

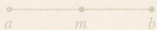
(It does converge — in fact to $\frac{3}{2} \ln 2$, while $s = \ln 2$.)

The Two Series, Side by Side





Riemann's Rearrangement Theorem



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.54: Riemann's Rearrangement Theorem

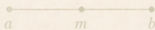
Theorem 3.54 (Riemann)

Let Σa_n be a series of *real* numbers which converges, but **not absolutely**.
Suppose

$$-\infty \leq \alpha \leq \beta \leq \infty.$$

Then there exists a rearrangement $\Sigma a'_n$ with partial sums s'_n such that

$$\liminf_{n \rightarrow \infty} s'_n = \alpha, \quad \limsup_{n \rightarrow \infty} s'_n = \beta. \quad (24)$$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

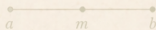
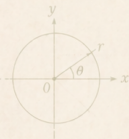
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

What the Theorem Gives Us

Conditional convergence: anything goes

For a conditionally convergent real series, rearrangements exist that:

- converge to **any prescribed sum** (take $\alpha = \beta =$ your number)



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

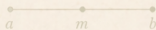
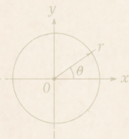
$$a^2 + b^2 = c^2$$

What the Theorem Gives Us

Conditional convergence: anything goes

For a conditionally convergent real series, rearrangements exist that:

- converge to **any prescribed sum** (take $\alpha = \beta = \text{your number}$)
- diverge to $+\infty$ or to $-\infty$ (take $\alpha = \beta = \pm\infty$)



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

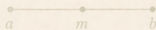
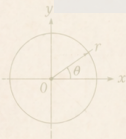
$$a^2 + b^2 = c^2$$

What the Theorem Gives Us

Conditional convergence: anything goes

For a conditionally convergent real series, rearrangements exist that:

- converge to **any prescribed sum** (take $\alpha = \beta = \text{your number}$)
- diverge to $+\infty$ or to $-\infty$ (take $\alpha = \beta = \pm\infty$)
- **oscillate** between any two prescribed bounds (take $\alpha < \beta$)



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof Strategy: Theorem 3.54

Goal: Rearrange $\sum a_n$ so that $\liminf s'_n = \alpha$ and $\limsup s'_n = \beta$.

Approach:

1. Split the series into its positive part and negative part; show **both diverge**.
2. Choose target sequences $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$.
3. **Greedy construction:** add positive terms until the sum exceeds β_1 ; add negative terms until it drops below α_1 ; repeat with $\beta_2, \alpha_2, \dots$
4. Overshoots are bounded by single terms, and the terms tend to 0 — so the turning points converge to β and α .

Proof of Theorem 3.54 — Step 1: Positive and Negative Parts

Step 1: Split a_n into positive and negative parts

For $n = 1, 2, 3, \dots$ put

$$p_n = \frac{|a_n| + a_n}{2}, \quad q_n = \frac{|a_n| - a_n}{2}.$$

Then

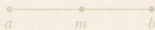
$$p_n - q_n = a_n, \quad p_n + q_n = |a_n|, \quad p_n \geq 0, \quad q_n \geq 0.$$

Proof of Theorem 3.54 — Step 1: Positive and Negative Parts

Step 1 (continued): $\sum p_n$ and $\sum q_n$ both diverge

- If *both* converged: $\sum(p_n + q_n) = \sum|a_n|$ would converge — but $\sum a_n$ is **not** absolutely convergent.
- If *exactly one* converged: from $\sum_{n=1}^N a_n = \sum_{n=1}^N p_n - \sum_{n=1}^N q_n$, $\sum a_n$ would diverge — but $\sum a_n$ **converges**.

Technique: proof by contradiction, twice — using *both* hypotheses.


$$a \quad m \quad b$$


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof of Theorem 3.54 — Step 2: The Two Reservoirs

Step 2: Collect the terms by sign

- $P_1, P_2, P_3, \dots$ = the **nonnegative** terms of Σa_n , in their original order
- $Q_1, Q_2, Q_3, \dots$ = the **absolute values of the negative** terms, in their original order

$\Sigma P_n, \Sigma Q_n$ differ from $\Sigma p_n, \Sigma q_n$ only by zero terms $\Rightarrow$ both **diverge**.

Proof of Theorem 3.54 — Step 3: The Greedy Construction

Step 3: Oscillate across moving targets

Choose real sequences $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$ with $\alpha_n < \beta_n$, $\beta_1 > 0$. Build

$$P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} + P_{m_1+1} + \cdots + P_{m_2} - Q_{k_1+1} - \cdots - Q_{k_2} + \cdots \quad (25)$$

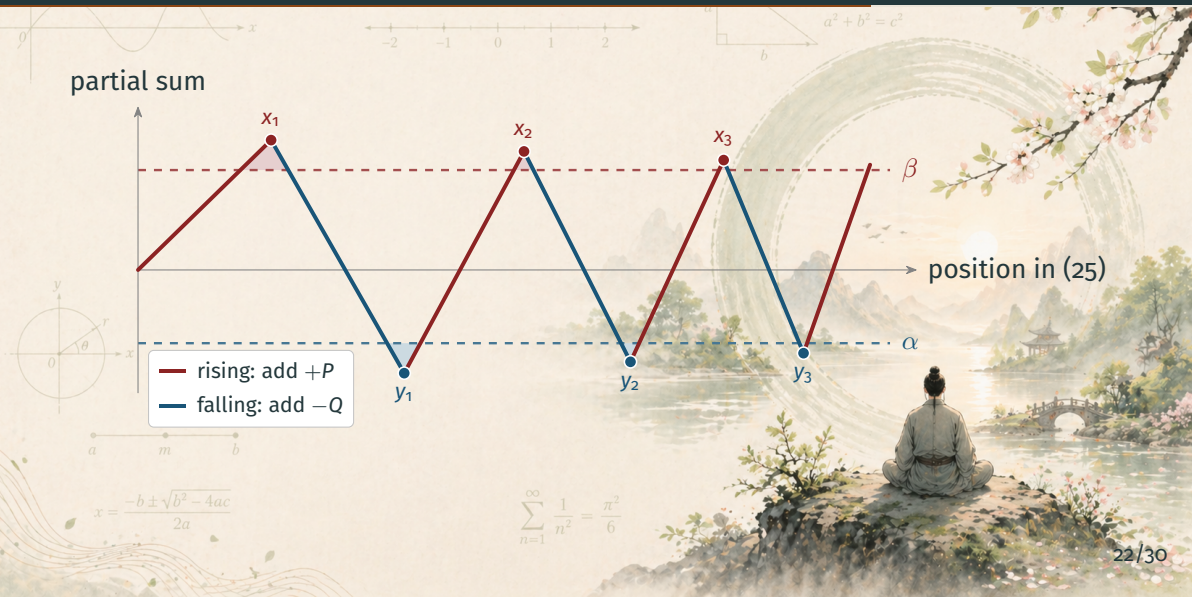
where $m_1, k_1, m_2, k_2, \dots$ are the **smallest** integers such that the partial sums successively

exceed β_1 , drop below α_1 , exceed β_2 , drop below α_2 , $\dots$

Possible because $\sum P_n, \sum Q_n$ **diverge**.

Clearly (25) is a rearrangement of $\sum a_n$.

Step 3 Visualized: Partial Sums Zigzag Across the Targets



Proof of Theorem 3.54 — Step 4: The Overshoots Vanish

Step 4: Turning points converge to β and α

Let x_n, y_n be the partial sums of (25) whose last terms are $P_{m_n}, -Q_{k_n}$.
Since m_n, k_n were chosen **smallest**:

$$|x_n - \beta_n| \leq P_{m_n}, \quad |y_n - \alpha_n| \leq Q_{k_n}.$$

But $\sum a_n$ converges $\Rightarrow a_n \rightarrow 0 \Rightarrow P_n \rightarrow 0, Q_n \rightarrow 0$, so

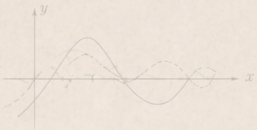
$$x_n \rightarrow \beta, \quad y_n \rightarrow \alpha.$$

Proof of Theorem 3.54 — Step 5: Conclusion

Step 5: No subsequential limit escapes $[\alpha, \beta]$

Between consecutive turning points, the partial sums of (25) move **monotonically** — so no number less than α or greater than β can be a subsequential limit of the partial sums.

$$\liminf_{n \rightarrow \infty} s'_n = \alpha, \quad \limsup_{n \rightarrow \infty} s'_n = \beta \quad (24)$$



$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Absolute Convergence and Rearrangements



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Theorem 3.55: Absolute Convergence and Rearrangements

Theorem 3.55

If $\sum a_n$ is a series of *complex* numbers which converges **absolutely**, then **every** rearrangement of $\sum a_n$ converges, and they all converge to the same sum.

Contrast:

Conditionally convergent $\Rightarrow$ anything goes (3.54).

Absolutely convergent $\Rightarrow$ **order is irrelevant**.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

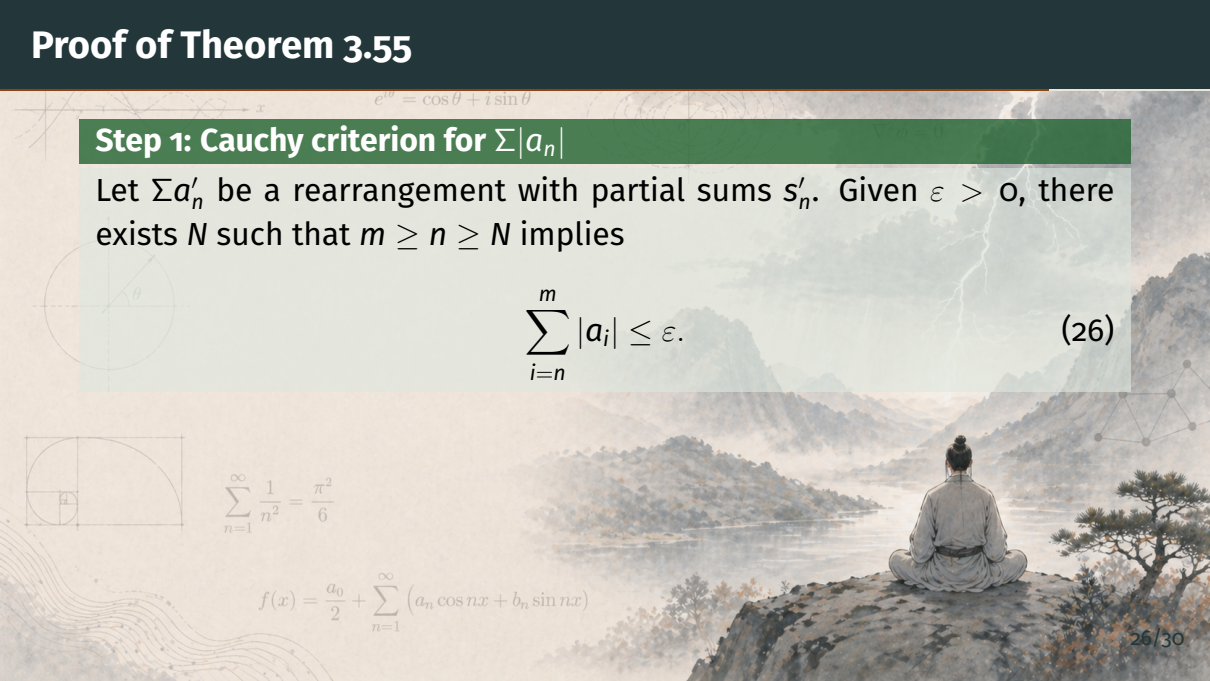
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.55

Step 1: Cauchy criterion for $\sum |a_n|$

Let $\sum a'_n$ be a rearrangement with partial sums s'_n . Given $\varepsilon > 0$, there exists N such that $m \geq n \geq N$ implies

$$\sum_{i=n}^m |a_i| \leq \varepsilon. \quad (26)$$


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.55

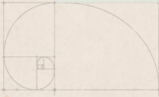
$$e^{i\theta} = \cos \theta + i \sin \theta$$

Step 2: The head cancels, the tail is small

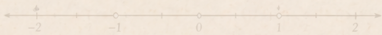
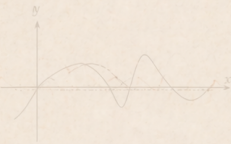
Choose p so that $1, 2, \dots, N$ are all contained in $k_1, k_2, \dots, k_p$. For $n > p$:

$$s_n - s'_n : \quad a_1, \dots, a_N \text{ cancel} \implies |s_n - s'_n| \leq \sum_{\substack{i > N \\ \text{finitely many} \\ \text{distinct}}} |a_i| \leq \varepsilon \quad \text{by (26).}$$

Hence $\{s'_n\}$ converges to the same sum as $\{s_n\}$. ■


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Summary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Rearrangement (3.52):** same terms, new order: $a'_n = a_{k_n}$, where $\{k_n\}$ is a permutation
- **Warning (3.53):** reordering alternating harmonic series **changes the sum**

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



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- **Warning (3.53):** reordering alternating harmonic series **changes the sum**
- **Riemann (3.54):** conditionally convergent real series: rearrangements achieve **any** $\liminf = \alpha$, $\limsup = \beta$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



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- **Warning (3.53):** reordering alternating harmonic series **changes the sum**
- **Riemann (3.54):** conditionally convergent real series: rearrangements achieve **any** $\liminf = \alpha$, $\limsup = \beta$
- **Absolute convergence (3.55):** **every** rearrangement converges to the **same sum** — order is irrelevant

Coming up next: Chapter 4 — *Continuity*.